

AIMLZ565 Machine Learning : A few worked examples

Support Vector Machines

Consider the following two-dimensional training dataset for a binary classification problem.

Point	(x_1, x_2)	y
1	(2, 2)	+1
2	(4, 4)	+1
3	(4, 0)	-1
4	(0, 0)	-1

Table 1: A two-dimensional small dataset

Using a linear Support Vector Machine:

- 1 Find the separating hyperplane.
- 2 Determine the support vectors.
- 3 Compute the margin.
- 4 Classify the point (3, 1).

Primal formulation

$$\begin{array}{ll}\text{minimize} & \Phi(\mathbf{w}) = \|\mathbf{w}\|^2 \\ \text{subject to} & y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 \quad 1 \leq i \leq N\end{array}$$

Primal formulation

$$\begin{array}{ll} \text{minimize} & \Phi(\mathbf{w}) = \|\mathbf{w}\|^2 \\ \text{subject to} & y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1 \quad 1 \leq i \leq N \end{array}$$

For the dataset in table 1, the linear separating hyperplane is of the form

$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ and the primal formulation is :

$$\begin{array}{ll} \text{minimize} & \Phi(\mathbf{w}) = \|\mathbf{w}\|^2 = w_1^2 + w_2^2 \\ \text{subject to} & \end{array}$$

$$1(2w_1 + 2w_2 + b) \geq 1$$

$$1(4w_1 + 4w_2 + b) \geq 1$$

$$-1(4w_1 + 0w_2 + b) \geq 1$$

$$-1(0w_1 + 0w_2 + b) \geq 1$$

Dual formulation

Hard margin dual formulation is :

$$\begin{aligned} & \text{maximize} && \mathbf{\Lambda} \cdot \mathbf{1} - \frac{1}{2} \mathbf{\Lambda} \cdot D \mathbf{\Lambda} \\ & \text{subject to} && \\ & && \mathbf{\Lambda} \cdot \mathbf{y} = 0 \\ & && \mathbf{\Lambda} \geq 0 \end{aligned}$$

where,

- $\mathbf{\Lambda}$ is a vector of Lagrange multipliers (one for each constraint) and $\mathbf{1}$ is a vector of ones of same length as $\mathbf{\Lambda}$
- D is a symmetric matrix where the $(i,j)^{th}$ entry is $D_{i,j} = y_i y_j \mathbf{x}_i \cdot \mathbf{x}_j$

The hard margin dual formulation for the dataset in table 1 is :

$$\mathbf{\Lambda} = [\lambda_1, \lambda_2, \lambda_3, \lambda_4]^T$$

$$D = \begin{bmatrix} 8 & 16 & -8 & 0 \\ 16 & 32 & -16 & 0 \\ -8 & -16 & 16 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

For instance,

$$D[1, 2] = y_1 y_2 \mathbf{x}_1 \cdot \mathbf{x}_2 = 1 \times 1 \times [2, 2] \cdot [4, 4] = 1 \times (2 \times 4 + 2 \times 4) = 16$$

Computing the decision boundary from the support vectors

The Support Vectors and corresponding Lagrange multipliers obtained are :

Support Vectors	Class Labels	Lagrange Multipliers
[4, 0]	-1	0.25
[0, 0]	-1	0.25
[2, 2]	1	0.5

$$\begin{aligned}
 \mathbf{w} &= \sum_{i=1}^l \lambda_i^* y_i \mathbf{x}_i \\
 &= -0.25 * [4, 0] - 0.25 * [0, 0] + 0.5 * [2, 2] = [0, 1]
 \end{aligned}$$

The equation of the decision surface for an arbitrary $\mathbf{x} = [x_1, x_2]^T$ is

$\mathbf{w} \cdot \mathbf{x} + b = 0$ i.e.

$$[0, 1] \cdot [x_1, x_2] + b = 0 \text{ i.e. } x_2 + b = 0$$

Calculating the bias term

Use complementary slackness condition for any support vector. i.e.

$$y_i(\mathbf{w} \cdot \mathbf{x}_i + b) = 1$$

Let us consider the support vector $[2, 2]$, this was the first example i.e. $\mathbf{x}_1$ and class label y_1 .

$$y_1(\mathbf{w} \cdot \mathbf{x}_1 + b) = 1 \text{ i.e. } 1 * 0.5 * ([0, 1] \cdot [2, 2] + b) = 1 \text{ i.e. } 1 + b = 0 \text{ i.e. } b = -1$$

EM algorithm for Gaussian Mixture Model

Problem Setup

We are given 1D data:

$$X = \{1, 2, 3, 10, 11, 12\}$$

Goal:

- Fit a Gaussian Mixture Model (GMM)
- Use:
 - $K = 2$ Gaussian components
 - EM algorithm

Assume:

$$\sigma_1^2 = \sigma_2^2 = 1$$

Initial parameters:

$$\mu_1^{(0)} = 2 \quad \mu_2^{(0)} = 11$$

$$\pi_1^{(0)} = 0.5 \quad \pi_2^{(0)} = 0.5$$

Gaussian Density Formula

For 1D Gaussian:

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Since:

$$\sigma^2 = 1$$

we get:

$$\mathcal{N}(x|\mu, 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2}\right)$$

Step 1: E-Step

Compute responsibilities:

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(x_n | \mu_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_n | \mu_j)}$$

Interpretation:

- Probability that point x_n belongs to cluster k

We compute responsibilities for each data point.

E-Step Example for $x = 1$

Compute Gaussian values.

For Cluster 1:

$$\begin{aligned}\mathcal{N}(1|2, 1) &= \frac{1}{\sqrt{2\pi}} e^{-(1-2)^2/2} \\ &= 0.3989 \times e^{-0.5} \approx 0.2420\end{aligned}$$

For Cluster 2:

$$\mathcal{N}(1|11, 1) = 0.3989 \times e^{-50} \approx 0$$

Now compute responsibility:

$$\begin{aligned}\gamma(z_{1,1}) &= \frac{0.5(0.2420)}{0.5(0.2420) + 0} \approx 1 \\ \gamma(z_{1,2}) &\approx 0\end{aligned}$$

Thus:

Responsibilities for All Points

Data Point	$\gamma(z_{n1})$	$\gamma(z_{n2})$
1	1.000	0.000
2	1.000	0.000
3	1.000	0.000
10	0.000	1.000
11	0.000	1.000
12	0.000	1.000

Observation:

- Left points belong to Cluster 1
- Right points belong to Cluster 2

Step 2: M-Step

Update means:

$$\mu_k = \frac{\sum_{n=1}^N \gamma(z_{nk}) x_n}{\sum_{n=1}^N \gamma(z_{nk})}$$

Update mixing coefficients:

$$\pi_k = \frac{N_k}{N}$$

where:

$$N_k = \sum_{n=1}^N \gamma(z_{nk})$$

Update Mean of Cluster 1

Responsibilities:

$$\gamma(z_{n1}) = [1, 1, 1, 0, 0, 0]$$

Compute:

$$N_1 = 1 + 1 + 1 = 3$$

Now update mean:

$$\begin{aligned}\mu_1 &= \frac{1(1) + 1(2) + 1(3)}{3} \\ &= \frac{6}{3} = 2\end{aligned}$$

Thus:

$$\boxed{\mu_1^{\text{new}} = 2}$$

Update Mean of Cluster 2

Responsibilities:

$$\gamma(z_{n2}) = [0, 0, 0, 1, 1, 1]$$

Compute:

$$N_2 = 3$$

Update mean:

$$\begin{aligned}\mu_2 &= \frac{10 + 11 + 12}{3} \\ &= 11\end{aligned}$$

Thus:

$$\boxed{\mu_2^{new} = 11}$$

Update Mixing Coefficients

Total number of samples:

$$N = 6$$

Update mixing coefficients:

$$\pi_1 = \frac{3}{6} = 0.5$$

$$\pi_2 = \frac{3}{6} = 0.5$$

Updated parameters:

$$\mu_1 = 2 \quad \mu_2 = 11$$

$$\pi_1 = 0.5 \quad \pi_2 = 0.5$$

EM Algorithm Summary

EM algorithm performed:

- ① E-Step:
 - Computed probabilities of cluster membership
- ② M-Step:
 - Updated cluster parameters

In this example:

- Clusters are well separated
- Responsibilities became nearly binary

In real datasets:

- Responsibilities are soft probabilities
- Multiple EM iterations are needed

EM Algorithm Summary

Repeat until convergence:

- 1 Initialize parameters

$$\mu_k, \sigma_k^2, \pi_k$$

- 2 E-Step:

$$\gamma(z_{nk}) = p(z_k | x_n)$$

- 3 M-Step:

- Update means
- Update variances
- Update mixing coefficients

- 4 Stop when likelihood converges

Key Insights

- GMM performs probabilistic clustering
- EM alternates between:
 - Estimating hidden variables
 - Updating parameters
- Unlike K-Means:
 - Assignments are soft
 - Clusters can have different variances

EM maximizes likelihood iteratively